

TG ECET (WP)-2026**SYLLABUS FOR METALLURGICAL ENGINEERING****MATHEMATICS (10 Marks)**

S.No	Syllabus
1	UNIT I: MATRICES AND DETERMINANTS Definition of Matrix-order of a matrix-Type of matrices - Algebra of matrices- sum, difference, scalar multiplication and product of 2x2 Matrix - Equality of two matrices-Transpose of a Matrix – Determinant of a square Matrix of order 2×2 .
2	UNIT II: CO-ORDINATE GEOMETRY Straight lines: Slope of straight line - Various forms of straight line: Point-Slope form, Two- Point form, Intercept form, and General equation of a Straight-line Conditions for Parallelism and Perpendicularity of the Straight lines – Equation of a straight line parallel and perpendicular to the given straight line -Distance between two Parallel lines.
3	UNIT III: TRIGONOMETRY Trigonometric Ratios of Compound angles (without proof) - Trigonometric Ratios of Multiple and Sub- multiple angles ($2A$, $3A$, $A/2$).

METALLURGICAL ENGINEERING (90 Marks)

S.No	Syllabus
1	UNIT I: MINERAL PROCESSING TECHNOLOGY: Introduction to metallurgy – ores & ore dressing, Methods of ore sampling – Comminution–sizing-classification, concentration - Minerals of commercially important metals.
2	UNIT II: FUELS & REFRACTORIES: Classification of solid, liquid and gaseous fuels – Testing and properties of important fuels- Manufacture and characteristics of Metallurgical Coke – Gasification of solid fuels - Combustion of fuels – Properties, manufacture and selection of Refractories.
3	UNIT III: METALLURGICAL THERMODYNAMICS: Introduction and applications of thermodynamics – First Law of thermodynamics - Thermo chemistry - Second Law of thermodynamics - Ellingham diagrams - Fugacity, activity and equilibrium constant - Phase equilibria – Solutions.
4	UNIT IV: PHYSICAL METALLURGY: Structure of Metals and Alloys – Solidification – Binary thermal equilibrium diagrams - Iron- carbon diagram - important non- ferrous binary alloy systems – Microscopic and macroscopic examination of metals and alloys.

5	UNIT V: HEAT TREATMENT TECHNOLOGY: Heat treatment of plain carbon steels - Annealing, Normalizing, Hardening and tempering of steels – TTT diagrams – Hardenability - Grain size, Quenching media. Alloy steels & Effect of alloying elements on plain carbon steels – Stainless steels, tool steels – Case hardening techniques. Special heat treatment techniques such as Austempering, Martempering, sub-zero treatment - Heat treatment of Non-ferrous metals and alloys – Age hardening – Heat treatment defects.
6	UNIT VI: FERROUS EXTRACTIVE METALLURGY: Iron ores and preparation of iron ores – Burden distribution – Blast furnace profile - Blast furnace plant and equipment – blast furnace reactions - irregularities and recent trends - sponge iron & methods of production – Ferroalloys – types and applications. Steel making by Bessemer, LD, Kaldo, Oxygen Lime (LDAC) process, Open hearth and Electric furnaces – Secondary steel making process – Vacuum treatment of liquid steel - Ingot defects - Continuous casting.
7	UNIT VII: NON-FERROUS EXTRACTIVE METALLURGY: Extraction of copper – Pyro and hydrometallurgical methods & refining – Aluminum - Extraction, Anode effect, Refining - Zinc and Lead - Pyro and hydrometallurgical extraction and refining. Extraction of Magnesium by Dows and pidgeon processes - Extraction of Titanium by Kroll's process - Refining of Titanium by Van arnell's process-Extraction of Thorium and Zirconium.
8	UNIT VIII: MATERIAL TESTING: Tension test. Stress - strain relationships, necking phenomenon. Hardness tests - principles and types. Impact testing - Notched bar impact tests. Transition temperature. Fatigue, Stress cycles, S-N diagram, Factors affecting Fatigue. Creep testing - creep curve, Stress - rupture test. Non- destructive testing- Principles, methods and applications of liquid penetrant, Radiography and Ultrasonic Magnetic particle.
9	UNIT IX: MECHANICAL METALLURGY: Plastic deformation of metals – lattice defects – Slip and Twinning - CRSS –Strengthening mechanisms. Strain hardening - Hot and Cold working - Recovery, recrystallisation and grain growth. Metal forming processes - Rolling, Forging, Extrusion & Sheet metal forming processes and defects.
10	UNIT X: FOUNDRY TECHNOLOGY: Patterns: Types, materials and pattern allowances, Moulding Sands - properties and Testing, Moulding Processes and equipment: Sand casting, Die casting, Shell moulding, Centrifugal casting, Investment casting and CO2 process-Cores: Types of Cores and properties, pouring and feeding of castings. Cast irons – types, Melting of cast irons - Grey, S.G and Malleable iron. Aluminum, Copper and Steel Foundry practices. Defects in Castings. Cleaning & Salvage of Castings.

11	UNIT XI: WELDING TECHNOLOGY: Basic concepts of Welding - Principles and processes of various welding techniques such as Oxy-acetylene, Shield Arc welding, Inert gas welding- TIG and MIG - Special welding processes - Plasma, resistance, electro slag, electron beam, thermit and Laser. Weld ability, factors affecting weld ability – Heat affected zone, Microstructure – Post weld treatments – Welding defects – Inspection and testing.